

PA1

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1 Question 1: Heuristic design and admissibility

1. Steps:

- (a) Num dif = 0 -->
For each letter in state -->
compare state letter with goal letter -->
if different -->
dif += 1
- (b) The unavoidable remaining cost is the cost that we aren't able to step directly to the answer via the number of differing letters. We might have to take a detour, an alternate word to work toward the answer. This lower bound doesn't account for that.
- (c) It cannot overestimate the remaining path cost because it assumes that you can go directly to the answer and not take detours. By only calculating the number of differing letters.

2. Scrabble:

- (a) Num totalCost = 0 -->
For each letter in state -->
If state letter differs from goal letter -->
totalCost += replacementLetter cost (replacement letter is the correct goal letter we want)
- (b) The unavoidable remaining cost is the cost that we aren't able to step directly to the answer via the number of differing letters and their corresponding cost. We might have to take a detour, an alternate word to work toward the answer which will increase the cost even more. This lower bound doesn't account for detours.
- (c) It cannot overestimate because it is assuming the optimal path exists and is what we want to take. As the optimal cost is each differing letter and its replacement letter cost. We don't account for detours, thus meaning we only calculate the shortest possible path.

3. Frequency:

- (a) totalCost = 0 -- >
differentLetters = 0
For each letter in state -- >
If state letter differs from goal letter -- >
different letters += 1
totalCost = different letters * (1 + minimum word value in dictionary)
- (b) The unavoidable remaining cost is that the most optimal path might not exist, but also we calculate our path assuming the cost is the smallest possible for a new word, but the reality is that the word might be less frequent (having a larger number), which would increase the cost to be even more.
- (c) It cannot overestimate the remaining cost, because it assumes we always pick the lowest cost word and has the most optimal path to completion. Which means that it cannot overestimate because the heuristic shows the lowest possible answer if that is even an answer, when in reality it might not be a valid path and that reality is a longer path. In other words, our path is only ever shorter than it actually is.

2 Question 2: Does the heuristic actually help?

- 1. They returned the same optimal cost.
- 2. A* produced the largest reduction in the steps cost measure.
- 3. For both word pairs the frequency heuristic was the least informative because when going from UCS using out heuristics to A*, we had the least amount of reduction, meaning that we had to do more work because the heuristic was less informative.
- 4. Steps heuristic was calculated based on the number of letter replacements needed. Which was direct and to the point, it gave the most information because it focused on getting to the answer as quick as possible.

Meanwhile Scrabble was the middle ground for providing information because it was steps but with a weight to certain letters, in favor of the more common ones, which pushed for keeping the more frequent letters to be the ones used, which would then in turn open up to more words.

The frequency heuristic would apply a weight to words and their frequency of being used. In this case we'd try to go for more frequent words, however because of this it was basically a roll of the dice for the cost of each word, which would provide the least amount of information, because it was the

least helpful in going to the goal and provided the least help in solving the case.

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
WHY → ASK	Steps	5	773	5	16	97.93%
WHY → ASK	Scrabble	9	756	9	342	54.7619%
WHY → ASK	Frequency	38.855	808	38.855	754	6.6832%
BEAR → DUCK	Steps	4	691	4	6	99.1317%
BEAR → DUCK	Scrabble	11	2952	11	1255	57.4864%
BEAR → DUCK	Frequency	38.240	1084	38.24	714	34.1328%

3 Question 3: Reversibility and direction-dependent costs

- Steps:
- 1 - yes
 - 2 - Since every action is reversible, and of equal cost the optimal path is still the optimal path for the reverse.
 - 3 - N/A

- Scrabble:
- 1 - no
 - 2 - This is because the starting and ending words are not equal, which means the replacement letters might not be of equal value, even though it is the same path of state, the cost is not equal.
 - 3 - The words WHY and ASK produce cost 14 and 9 depending on the way in which you go.

- Frequency:
- 1 - no
 - 2 - This is because the minimum words are not equal in value, and the heuristic is (num of dif letters times the min cost word), thus creating a difference in what the cost might be.
 - 3 - The words CAT and DOG produce cost 23.02 and 21.96 depending on the way in which you go.

4 Question 4: LLM heuristic audit

1. **AI:** Gemini

Model: Flash-Lite

Prompt: In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary. The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal.

Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.”

2. : **AI Model Heuristic:**

We define the rarity-aware heuristic $h(n)$ as:

$$h(n) = \begin{cases} 0 & \text{if } n = \text{goal} \\ h_0(n) + \min_{w \in N(n)} \text{rarity}(w) & \text{if } n \neq \text{goal} \end{cases}$$

Where:

- $h_0(n)$ is the baseline heuristic (Hamming distance: the number of differing character positions between n and goal).
 - $\min_{w \in N(n)} \text{rarity}(w)$ is the minimum rarity value among all valid single-step dictionary neighbors of n .
3. This heuristic is the cheapest because it will always add the number 1 for every letter, which never changes, and we know based off of question 3 that it is also reservable (this is the steps heuristic) and still admissible. Also because this only adds the cheapest option for the next move and not further options, it has to be admissible because it only factors in the cheapest possible paths, hence it has to be admissible.
 4. The proposal is valid.
 5. It was the exact same, although the formula was different by where the * and + was, the exact numbers ended up being the same because the mathematical properties that apply actually make the 2 equations equal. In other words, the equation provided by the AI model actually translates to what I have even though they are written differently.

AI Use Disclosure: AI was used in helping write stuff in latex code, as I had had issues with writing some math and making charts. I also used AI for Question 2 in generating the terminal commands to fill out the information in the chart, however I did then comeback and make the changes as stated in your announcement and redo my work, which all that did was verify my already done work was actually correct.